Using hotkeys, we can do this directly by ctrl-num, where num is the number of subdivisions we want in view mode.

By default, the subdivisions are in Catmull Clark subdivisions, which distorts the mesh as it subdivides. If we want a subdivision that does not change the shape of the object we can go to the tab Simple subdivisions and work.

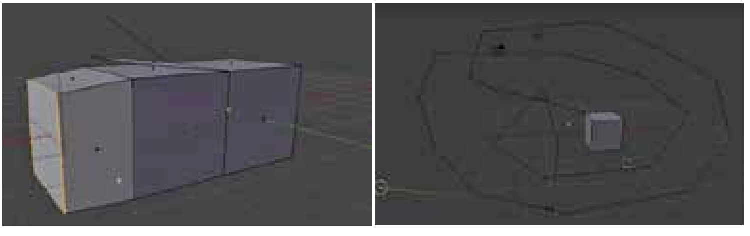
Extrude tool

The extrude tool can be used to extrude or extend a vertices, edge or face of a particular object in edit mode. By using hotkey E, and then selecting the vertices/ edge/face of the object and extruding it in a particular direction.

As with previous transformations, we can left-click to confirm or right-click to cancel. However, right clicking does not delete the extrusion. Instead, it saves the extrusion but brings the moved vertice/edge/face back to its original position. So if we use the G tool, we can see that the vertice is movable. This is useful in scaling, as we can scale and extrude to make objects of different shapes. The extrusion by default happens along the normal, to make the extrusion happen freely we can press Z, which makes the object move in all directions, and then press a particular axis to make it move along the global X, Y or Z axis. Double tapping an axis key will make it move along axes local to the object.

Another way to extrude is by using ctrl-left click. This extrudes the vertice to wherever the mouse cursor is so that we can make irregular shapes fast and efficiently. However, here we can't work along with an axis.

Another function is single extrusion (in tools), which makes each selection extrude along its own normal, which is particularly helpful in shapes like spheres, where each face has a different normal. We can extrude several faces along their own normal without scaling them or making them all move in one direction.



Extrusion by Hotkey E and by Ctrl-left click